



Weapons of Influence Misused: A Social Influence Analysis of Why People Fall Prey to Internet Scams¹

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Abstract

Internet crime, including fraud and spread of malicious software, is a pervasive and costly global issue. Many of these crimes occur not because of technology failure but because of the human element. People can easily be manipulated through social engineering – the use of psychological tactics to influence individuals to assist in their own victimization. We employ a social influence framework, drawing upon ideas from Robert Cialdini, to understand the nature of social engineering techniques and why they are successful in exploiting unsuspecting individuals. Specifically, we discuss how social engineers misuse six pervasive weapons of influence – *Liking, Authority, Scarcity, Social Proof, Reciprocity, and Commitment and Consistency* – in order to effectively influence individuals to take the initial steps toward their exploitation. We conclude with a discussion of the precautions individuals can take to resist malicious influence attempts online.

Imagine a friend contacts you via email and says he is stranded in a foreign country because he was robbed. He implores you to help him get home by sending money via a Weblink provided within the email and promises to pay you back when he returns home. What would you do? Would you be like many people and click on the link to help your friend out? Research indicates that most people would at least be sympathetic, especially regarding a friend, and would click on the link. In fact, this is a popular email scam known as the “Stranded Traveler” (Leamy & Hawkins, 2012, July 13). This type of Internet scam is manufactured by social engineers whose main goal is to exploit Internet users by getting them to take the first step(s) in their own victimization. Unfortunately, those who took the first step by clicking on the link may have inadvertently provided others with access to their personal information and money.

The purpose of this paper is to utilize a social influence framework to (1) understand why social engineering scams such as in the previous example are successful and how they work and (2) inform individuals about this global issue and how social psychological tactics can be utilized for deceptive purposes. First, we briefly define social engineering and social influence. Then, we demonstrate how social engineers misuse each of Cialdini’s (2009) social influence principles: *Liking, Authority, Scarcity, Social Proof, Reciprocity, and Commitment and Consistency*. We do this by examining real life examples of common Internet scams while highlighting the social influence tactic utilized. While social engineers can exploit individuals using a variety of tactics, including directly hijacking a computer without the user knowing, we specifically focus on scams that attempt to influence the user to take the first steps in their own exploitation (i.e., by clicking on a link or providing personal information to others) and whose primary goal is to gain reward or access to private information. Finally, we discuss some general preventative methods that individuals can take.

What is Social Engineering on the Internet?

In an online context, social engineering is a means of gaining access to systems or data by exploiting human psychology, as opposed to using direct technical hacking methods (Goodchild, 2010, January 11). On the Internet, social engineering scams are prominent because they often are efficient (i.e., the user does some of the work and the method of delivery is free). Because of the ease of sending and receiving information online, social engineers have the ability to reach a large number of potential victims. Internet crimes, including social engineering attempts online are prevalent and have been steadily increasing along with large monetary losses (Atkins & Huang, 2013; Internet Crime Complaint Center, 2013a, May 5). For instance, in 2012, almost 290,000 Internet crimes were reported in the United States (Internet Crime Complaint Center, 2013a, May 5). Further, some sources have estimated the worldwide monetary costs of such crimes to be in the 100 to 200 billion USD range (Chang, Ku, Wu, & Chiu, 2012; Kshetri, 2013).

One of the most common ways people are exploited on the Internet is through phishing scams. Phishing is a technique used to gain access to personal information and money (Atkins & Huang, 2013; Larcom & Elbirt, 2006). Most commonly, as in the “Stranded Traveler” technique, this occurs when a target of an Internet scam receives a seemingly legitimate email that requests personal information or money, either directly in the email or by visiting another legitimate-sounding link. Usually, after clicking on the link, the user is directed to a fraudulent site where personal information is entered (e.g., bank account, credit card information, and social security number) and this can result in identity theft (McMillan, 2006, April 3). Another common social engineering tactic involves malicious software (also referred to as malware). Malware is installed on a user’s computer and performs unwanted tasks, such as running adware – popup advertising (Abraham & Chengalur-Smith, 2010). Malware can be used to illegally gain profit because many programs capture personal information, such as passwords from a person’s computer, which may then be used to access financial accounts. While malware can be directly installed (i.e., by a hacker), it is becoming more common for social engineers to design malware that individuals inadvertently install themselves (i.e., by clicking a link). In the next section, we discuss some common phishing and malware scams that have been successfully employed by utilizing social influence tactics. First, we briefly define social influence and why individuals may be more susceptible to influence attempts online. Then, we provide examples of Internet scams that utilize each of the six influence principles to increase their effectiveness.

Social Influence: Weapons Misused

Social influence is a change in a person’s attitudes, beliefs, or behaviors that results from real or imagined pressure(s) (Cialdini, 2009). Cialdini refers to six universal social influence principles or tactics, which can be misused as “weapons of influence”: *Liking, Authority, Scarcity, Social Proof, Reciprocity, and Commitment and Consistency*. These are commonly employed to persuade individuals or gain compliance in everyday face-to-face settings. The main reason these principles are effective is because they serve as heuristics (mental shortcuts) – a mode of information processing that is commonly relied upon because people are cognitive misers. That is, individuals do not always have the ability or motivation to engage in very effortful and careful thinking (Fiske & Taylor, 2013). Furthermore, this may be particularly true for many online interactions (Guadagno, Okdie, & Muscanell, 2013).

We argue that the ease of Internet connectivity has led to some complacency among many Internet users and thus makes them more susceptible to social influence tactics. It has become commonplace to utilize the Web for fulfilling everyday purposes. With the click of a button,

individuals can manage their personal finances and documents containing personal information, and they can engage in social interactions and other leisure activities (e.g., email, online social networking, games, and shopping), which may involve exchange of personal information. This reliance on the Web in everyday life may lead individuals to feel less concerned about being taken advantage of online. Indeed, some research supports this notion, showing that many people hold a self-serving bias (“this can’t happen to me”) and think they (but not others) are immune to being exploited online (Debatin, Lovejoy, Horn, & Hughes, 2009). Additionally, we contend that being constantly connected to the Internet and having easy access to an unlimited amount of information may lead to heuristic processing in online contexts (Guadagno, Okdie, & Muscanell, 2013). Because the Internet offers so much information, individuals can experience cognitive overload and may be more likely to rely on heuristics such as social influence principles, as opposed to thinking effortfully. Some research has already shown that individuals routinely rely on heuristic cues when judging information and making decisions online (Guadagno & Cialdini, 2005; Guadagno, Muscanell, Rice, & Roberts, 2013; Metzger, Flanagin, & Medders, 2010). Additionally, individuals often multitask online which also increases cognitive load and decreases effortful thinking (Srivastava, 2013). Given that individuals may be more complacent and prone to thinking heuristically online, there may be a greater risk of exploitation. Next, we discuss each of the six social influence principles as they are utilized within social engineering scams online.

Liking

The principle of liking involves a decision heuristic that makes likeable people more influential. The idea is that likeable people are usually more familiar, similar, and trustworthy as compared to unlikable others and thus people use a “likeable = good source of information” decision heuristic when evaluating evidence from a likeable source (Cialdini, 2009). Research supports this assumption, for example showing that more participants complied with a request, purchasing raffle tickets, from a likeable requestor as compared to an unlikeable requestor (Regan, 1971). The effects also extend to online settings – individuals demonstrate more attitude change (Guadagno & Cialdini, 2002, 2005, 2007) and changes in behavioral intentions when communicating with someone they like (Holzwarth, Janiszewski, & Neumann, 2006). Liking can be easily manipulated and misused by social engineers online. For example, in the case of the phishing scam known as the “Stranded Traveler,” an email appears to be sent by a friend or known contact. The social engineer obtains information about a user’s contacts and then constructs an email that utilizes the contact’s email address. This tricks the message recipient into thinking they are being contacted by a friend or at least someone they know. The liking heuristic increases the chance that individuals will take the first steps in this type of scam. First, individuals are more likely to open the email simply because it comes from a friend – someone they like and probably trust. For this reason, they will be more likely to comply with the request from the fictitious friend and click on the link. It is unlikely that these types of influence attempts would be as successful when coming from an unknown person with whom influence targets have had no opportunity to develop liking for.

Similar scams capitalize on liking via social media. In a newer scam, individuals are sent a friend request on Facebook that is seemingly from an old friend (Better Business Bureau, 2013, July 1). Once the request is accepted, the scammer initiates conversation and introduces information about eligibility for a federal grant. The link however, is a fake link, and the goal is to gain personal information from users who visit the page (i.e., to later aid in identity theft). One person who accepted the phony friend request explained, “I listened simply because she is my friend.”

Not all Internet scams utilize a “phony friend” technique. Some social engineers also manipulate liking online by trying to increase their own likability and not by trying to pretend to be a known other. For example, in a recent Valentine’s Day scam, individuals are contacted on an online dating or social networking site (Halper, 2014, February 12). The social engineer pretends to be romantically interested in the user and then spends several weeks or so forming a connection with him/her. By sharing conversations and feigning similar interests, the engineer is able to establish his/her own likability in the eyes of the potential victim. Later, after becoming liked and trusted by their target, she/he asks for money, which was the ulterior motive all along. By utilizing cues associated with liking, social engineers can increase the chance that individuals will at least take the first step in being exploited (i.e., opening and reading an email or accepting a chat request).

Authority

The principle of authority indicates that authorities know better than others. Individuals often see authority figures as being people who are credible, have expertise, and/or have direct power. Thus, people find their opinions to be more valuable, and they are more likely to be influenced by authorities (Cialdini, 2009). Research has demonstrated the powerful effects of authority as a social influence principle (Milgram, 1963). Sometimes, mere trappings of authority (e.g., a lab coat) are extremely influential. This is often a useful heuristic, as it generally makes sense to seek and trust the opinions of medical doctors, scientists, political figures, and the like. However, similar to the principle of liking, authority can easily be misused by social engineers online. For example, in 2012, there was a phishing scam in which emails were sent to US Department of Defense military members and were seemingly from the Defense Finance and Accounting Services (Internal Revenue Services, 2012, June). These emails asked individuals to send various VA and IRS documents containing personal information in order to receive compensation they were eligible for. The social engineers utilized cues of authority by sending the emails from a “.mil” email address (a military address). It is likely that many of the victims assumed that because the email came from an official government department and from an apparently credible email address, they should comply. More recently, a variety of spam emails were reported, which contain important Federal Bureau of Investigation (FBI) officials’ names (Internet Crime Complaint Center, 2013b, September 25). In some of the emails, it appears that a high ranking official, the director of the FBI, is requesting money. By using names and affiliations associated with those in authority positions, social engineers can trick people into complying because their inclination is to do what the authority says.

Scarcity

The principle of scarcity is that scarce objects or opportunities are valuable. Individuals attach more value when the object or opportunity in question is perceived to be difficult to obtain or limited in supply. For example, in one study, buyers for supermarkets doubled their beef orders when they were told that that particular beef was scarce (Knishinsky, 1982). Furthermore, when the buyers were told that the information itself was scarce, they ordered six times more beef. In some cases, this heuristic is valid because scarce resources can indeed be more valuable (e.g., high quality products sell quickly and become scarce). However, social engineers can misuse this principle to entice individuals to click on a Weblink or download software. The Koobface malware (virus) is a primary example, and it has been a persistent problem for Facebook users (Slattery, 2008, December 5). Koobface is engineered to contain a link to “secret” or hard to find media (Goodchild, 2009a, July 13). By making

access to this media appear to be limited, it encourages users to click on the link because this makes it suddenly seem more desirable. Once users click on the link and/or download the software, the virus installs, and the users' computer becomes a botnet, an automated computer program, which can aid in the theft of personal information. Other popular scams that utilize the principle of scarcity include "limited time only" offers. For instance, in 2011, links were sent out via emails and Facebook purporting to offer \$100 Walmart gift cards (Spilewsky & Healy, 2012, April 25). Many were accompanied by messages saying "For a limited time only". Thus, social engineers can falsely present resources as being scarce in order to increase their perceived value, making their attempts more effective.

Social proof

The principle of social proof indicates that people should do what others are doing. Individuals often look to others' behavior, especially if there are many others engaging in a behavior and also if the situation is ambiguous. In a classic study, a hotel was able to increase guests' towel recycling behavior by 25% by providing social proof information (Goldstein, Cialdina, & Griskevicius, 2008). When guests believed that other patrons also recycled their towels, they were more likely to do so as compared to individuals who received only a simple request to recycle their towels. People often rely on social proof as a heuristic because many times it is useful to follow the behavior of others (e.g., as a newcomer in an organization, you look to other members to learn what to do). This tactic may be observed in data harvesting applications on Facebook. When users notice that many of their friends have installed an application, they do so too assuming that the application is fun or useful without thought as to its potential for compromising their privacy (Goodchild, 2009b, August 31). More recently, social proof has been misused by social engineers in the form of fake "liking" or fake "fans" via social media (Finkle, 2013, August 16). That is, computer programs are now being developed to generate fake "likes" or followers/fans on sites like Facebook, Twitter, Instagram, and YouTube. Generating fake support serves to create the illusion of social proof (having a lot of interest and support) and in many cases, to influence individuals to spend money. Thus, individuals may be more likely to visit a page, click on related links, and to "like" or "follow" the page if they see that supposedly many others are doing the same. Sometimes, this may not be an overt malicious influence attempt (i.e., the engineer is not directly stealing your personal information), but other times, this can allow others to gain access to sensitive information that may be utilized maliciously. By falsely creating the appearance that many other people clicked on a link or liked a page, more individuals can be influenced to do the same because they often look to the behaviors of others without thinking very carefully about it.

Reciprocity

The principle of reciprocity can be summed up by the Golden Rule, "do unto others as you would have them do to you." In other words, when someone does you an act of kindness, it is an assumption that at some point, you should reciprocate. Thus, individuals are more likely to comply when they feel indebted to one another. A classic demonstration of reciprocity is the door-in-the-face technique (Cialdini et al., 1975). Participants were more likely to comply with a request if they were first presented with an exceptionally large request (one they would say no to, i.e., "Will you volunteer to be a counselor twice a week for two years?") and then a second smaller request (i.e., "Will you chaperone a group of juveniles on one two hour trip?"). People comply with the second request, because they feel they owe the

requestor who did them a favor by conceding from an unreasonable request to something more reasonable. This heuristic is useful because reciprocity helps to maintain relationships. However, the principle of reciprocity can be misused in online contexts. In fact, various forms of wire-back scams, such as the infamous “Nigerian Prince” scam, utilize the reciprocity principle (Tompore, 2013, September 21). In these scams, the social engineer does an unsuspecting user a favor up front (by sending a check or by paying too much for something up front – before ever receiving anything themselves). Afterwards, the social engineer will ask that person to wire some money back. In reality, the money that was originally sent is usually in the form of a fraudulent check that would bounce soon, causing the target to lose money. By doing the kindness of sending money up front (without having received anything themselves), the social engineer invokes the principle of reciprocity, and the target of the attempt then feels obliged to comply because they owe that person.

Commitment and consistency

The principle of commitment and consistency is that being consistent and staying true to commitments is good. In general, individuals like to be reliable, stable, and dependable individuals. Inconsistencies in the way a person thinks and behaves can alert to others that one is not very reliable or stable. Thus, individuals often behave in ways that appear to be consistent with previous behaviors. A classic example of this is the foot-in-the-door technique, similar to the door-in-the-face, except that it works when a small request is presented before the larger request (Burger, 1999). For instance, when women agreed to answer some questions on the telephone about household products they use, they were then more likely to later allow a team to come to their homes and survey their household products for 2 h (Freedman & Fraser, 1966). The idea is that once people agree to a small commitment (which they are likely to say yes to), they will also say yes to subsequent, larger requests in order to maintain consistency in their behaviors. This principle can be seen in the “Stranded Traveler” example, in which the target receives a request that appears to be from a friend who is stuck abroad without money. Scams such as this, and more generally ones in which an initial request appears to come from a known other can then often escalate, for instance by getting the unsuspecting user to engage in further conversation and gradually asking the user to send personal information and even money. By getting unsuspecting individuals to make an initial commitment (accepting a friend request), it increases the likelihood that a social engineer will then be successful in subsequent attempts (asking and receiving personal information). Many common Internet scams utilize the principle of commitment and consistency, as the goal is often to persuade individuals to take a smaller initial step, something most people would say yes to (opening and reading an email from a friend or authority figure). If people commit to the first step, they may be more likely to further aid in their own exploitation online because they want to continue to be (and be seen as) consistent.

Best Practices for Resisting Social Influence Attempts on the Internet

While social engineers can misuse any combination of social influence tactics via the Internet, it is possible to teach individuals to be more resistant to such attempts (Sagarin & Mitnick, 2012). Specifically, Sagarin and Mitnick propose that teaching individuals to realize that they are vulnerable to influence attacks online (i.e., “this can happen to me”) and getting them to understand what online behaviors will lead to negative consequences (e.g., filling in information or clicking “like” on social media can allow others to gain their personal information) can decrease their likelihood of being taken advantage of. Cialdini (2009) also proposes preventative methods for each of the six principles of social influence. Mainly, the best lines

of defense involve being informed about their existence and how they work and also engaging in effortful thinking as opposed to using mental shortcuts online. Below, we briefly present some of Cialdini's cognitive strategies that may be used to resist weapons of influence.

Liking. The best defense against unethical use of the liking principle is to try to separate the likeable person from their influence attempt upon recognizing that they are likeable. That is, individuals should try to think about that person's request in isolation, keeping in mind that likeability might play a factor in their decision. A good question to consider might be *Would I say yes to this request if someone else were asking me?*

Authority. The best defense against malicious use of the authority principle is to ask two specific questions before deciding how to act: (1) *Is this person truly an expert?* (2) *How truthful do I expect this person to be based on his or her position?* By considering these questions, individuals may be less likely to be influenced by trappings of authority (e.g., clothing) and able to think more carefully about the supposed authority's validity and credibility.

Scarcity. The best defense against the scarcity principle is to recognize (and anticipate) an experience of increased physical arousal that results from finding out something is limited in availability. Anticipating this arousal could give individuals the chance to calm themselves down and create more time to think carefully once that "rush" passes. It may also be helpful to keep in mind that scarce does not always mean valuable or good.

Social proof. The best defense against the misuse of the social proof principle is to understand that other peoples' actions can sometimes be wrong and under which circumstances. In general, individuals should remember that many people engaging in a behavior does not equate to it being good or correct (e.g., riots and mob behavior). Also, similar to the defense against the liking principle, it may be best to isolate an influence attempt from others' behaviors.

Reciprocity. The best defense against the malicious use of the reciprocity principle is to reject the rule (i.e., by refusing to comply with the initial request). Recognizing that a favor was given as a means of manipulation reasons is the best way to accomplish this. This may be difficult to accomplish, but individuals should try to think carefully about what the requestor's true motives are (*Is it likely this person really did me this kindness without expecting anything in return?*). By recognizing misuse, individuals can reject subsequent requests because they no longer fit the principle of reciprocity.

Commitment and consistency. The best defense against the misuse of the commitment and consistency principle is twofold: First, individuals need to recognize when they are being influenced to do something that they do not want to do (the feeling of not wanting to comply may be more likely to occur when presented with subsequent larger requests). Second, *before acting*, individuals should think about what they would do in hindsight (*If I could go back would I do the same thing?*). By asking this question before committing to a response, it increases the chance of recognizing that this principle is being misused and decreases the likelihood of following up with subsequent commitments.

Conclusions

Owing to its ease of use and widespread accessibility, the Internet is here to stay. Individuals will continue to be exposed to myriad influence attempts, including ones from social engineers with malicious intentions. Because Internet use has become commonplace, individuals may be more complacent in their online information processing and decision making.

Further, constant connectivity and the tendency to multitask online may increase cognitive load leading people to rely on heuristics, including social influence principles, making them more vulnerable to malicious attempts online. Based on suggestions from both Sagarin and Mitnick (2012) and Cialdini (2009), the best defense that Internet users have to keep them from becoming victims of social engineering tactics is to be aware that they are not invulnerable, to be informed about the social influence tactics and their mechanisms, and for the most part, to think more effortfully and analytically about persuasive messages.

Short Biographies

Dr Nicole Muscanell is a postdoctoral researcher at the Knowledge Media Research Center in Tübingen, Germany. She received her PhD in Social Psychology from the University of Alabama. Her research examines social influence, self-processes, and individual differences as they pertain to online communication.

Dr Rosanna Guadagno is a program officer at the National Science Foundation. She received her PhD in Social Psychology from Arizona State University, completed her postdoc at the University of California, Santa Barbara, and was on the faculty at the University of Alabama. Her research examines social influence and persuasion, technology-mediated social interaction, and gender roles.

Shannon Murphy, MA is a doctoral student in Social Psychology at the University of Alabama. Her research examines social psychological aspects of health behavior and outcomes, evaluations of education and community interventions, and social influence in online settings.

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¹ The views expressed in this manuscript do not reflect those of the National Science Foundation.

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